P9.19 $\hat{H} = \frac{\hat{L}^2}{2I}$

$$\langle \theta, \phi | \psi(0) \rangle = \sqrt{\frac{3}{4\pi}} \sin\theta \sin\phi$$

(a) Comparing with (9.152a) we see that

$$\langle \theta, \phi | \psi(0) \rangle = \frac{i}{\sqrt{2}} \left[Y_{1,1}(\theta,\phi) + Y_{1,-1}(\theta,\phi) \right]$$

In terms of kets alone

$$|\psi(0)\rangle = \frac{i}{\sqrt{2}} |1,1\rangle + \frac{i}{\sqrt{2}} |1,-1\rangle$$

The state evolves as

$$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar} |\psi(0)\rangle = e^{-\frac{it}{\hbar}\frac{\hat{\vec{L}}^2}{2I}} \frac{i}{\sqrt{2}} \left(|1,1\rangle + |1,-1\rangle \right)$$

$$= \left(\hat{L}^2 |1,\pm 1\rangle = 1(1+1)\hbar^2 |1,\pm 1\rangle = 2\hbar^2 |1,\pm 1\rangle \right) =$$

$$= e^{-i\frac{\hbar t}{I}} \frac{i}{\sqrt{2}} \left(|1,1\rangle + |1,-1\rangle \right)$$

$\Rightarrow$ the state picks up a phase:

$$\langle \theta, \phi | \psi(t) \rangle = e^{-i\hbar t/I} \sqrt{\frac{3}{4\pi}} \sin\theta \sin\phi$$

(b) Since $|\psi(0)\rangle = \frac{i}{\sqrt{2}} |1,1\rangle + \frac{i}{\sqrt{2}} |1,-1\rangle$

we'll have $L_z = \pm\hbar$

with probability $1/2$.

(c) $\langle L_x \rangle = ?$ $\quad \hat{L}_x = \frac{1}{2}(\hat{L}_+ + \hat{L}_-)$

$$\Rightarrow \langle L_x \rangle = \frac{1}{2} \langle \psi(t) | \hat{L}_+ + \hat{L}_- | \psi(t) \rangle =$$

$$= \frac{1}{2} \langle \psi(t) | e^{-it\hbar/I} \frac{i}{\sqrt{2}} \left(\sqrt{2}\hbar |1,0\rangle + \sqrt{2}\hbar |1,0\rangle \right)$$

$$= \hbar \left(+ \frac{1}{\sqrt{2}} \right) \left(\langle 1,1| + \langle 1,-1| \right) |1,0\rangle = 0.$$

(d) This is tricky question.

We know that in principle L_x can be $\pm\hbar$ or 0.
$(m = +1, 0, -1)$

From $\langle \hat{L}_x \rangle = 0$ in part (c) we deduce that state $|\psi(t)\rangle$ has equal amplitudes for $\hat{L}_x = \pm\hbar$ [for otherwise we could not have $\langle \hat{L}_x \rangle = 0$].

Thus $|\psi(t)\rangle = a\,|1,1\rangle_x + b\,|1,0\rangle_x + a\,|1,-1\rangle_x$

these are states in L_x-basis

Note that this implies

$$\langle \hat{L}_x \rangle = \left({}_x\langle 1,1|\, a^* + b\,{}_x\langle 1,0| + a^*\,{}_x\langle 1,-1| \right)\left(a\hbar|1,1\rangle_x - a\hbar|1,-1\rangle_x \right)$$

$$= |a|^2\hbar - |a|^2\hbar = 0 \quad \text{as it should be.}$$

Now recall <u>P3.15</u> : $|1,0\rangle_x = \frac{1}{\sqrt{2}}|1,1\rangle - \frac{1}{\sqrt{2}}|1,-1\rangle$

$$\Rightarrow b = {}_x\langle 1,0|\psi(t)\rangle = \left(\frac{1}{\sqrt{2}}\langle 1,1| - \frac{1}{\sqrt{2}}\langle 1,-1| \right) \cdot e^{-it\hbar/I} \cdot$$

$$\cdot \frac{i}{\sqrt{2}}\left(|1,1\rangle + |1,-1\rangle \right) = i\,e^{-it\hbar/I}\left(\frac{1}{2} - \frac{1}{2} \right) = 0$$

$\Rightarrow b = 0 \Rightarrow$ from normalization $(2|a|^2 + |b|^2 = 1)$ we find $a = \pm\frac{1}{\sqrt{2}}$

$\Rightarrow$ the measurement of $\hat{L}_x$ would result in $\pm\hbar$ with probability 50% each.

P9.20

$$\hat{H} = \frac{\hat{L}^2}{2I} + \omega_0 \hat{L}_z$$

$$\langle \theta,\phi | \psi(0) \rangle = \sqrt{\frac{3}{4\pi}} \sin\theta \sin\phi = \frac{i}{\sqrt{2}} \left[Y_{1,1}(\theta,\phi) + Y_{1,-1}(\theta,\phi) \right]$$

Now $\begin{cases} \hat{L}^2 \, Y_{\ell,m}(\theta,\phi) = \hbar^2 \ell(\ell+1) \, Y_{\ell,m}(\theta,\phi) \\ \hat{L}_z \, Y_{\ell,m}(\theta,\phi) = \hbar m \, Y_{\ell,m}(\theta,\phi) \end{cases}$

$$\Rightarrow \hat{H} \, Y_{\ell,m} = \left[\frac{\hbar^2 \ell(\ell+1)}{2I} + \hbar\omega_0 m \right] Y_{\ell,m}$$

$$\Rightarrow \langle \theta,\phi | \psi(t) \rangle = \langle \theta,\phi | e^{-i\hat{H}t/\hbar} | \psi(0) \rangle =$$

$$= \frac{i}{\sqrt{2}} \left[e^{-i\frac{\ell(\ell+1)\hbar}{2I}t - i\omega_0 t} \, Y_{1,1} + e^{-i\frac{\ell(\ell+1)\hbar}{2I}t + i\omega_0 t} \, Y_{1,-1} \right]$$

$$= (\ell=1) = \frac{i}{\sqrt{2}} e^{-i\frac{t\hbar}{I}} \left(e^{-i\omega_0 t} Y_{1,1} + e^{i\omega_0 t} Y_{1,-1} \right)$$

$$= \frac{i}{\sqrt{2}} e^{-i\hbar t/I} \sqrt{\frac{3}{8\pi}} \sin\theta \underbrace{\left(-e^{i\phi - i\omega_0 t} + e^{-i\phi + i\omega_0 t} \right)}_{2i \sin(-\phi + \omega_0 t) = -2i \sin(\phi - \omega_0 t)}$$

$$= \sqrt{\frac{3}{4\pi}} e^{-i\hbar t/I} \sin\theta \sin(\phi - \omega_0 t)$$

$$\langle \hat{L}_x \rangle = \langle \psi(t) | \tfrac{1}{2}(\hat{L}_+ + \hat{L}_-) | \psi(t) \rangle =$$

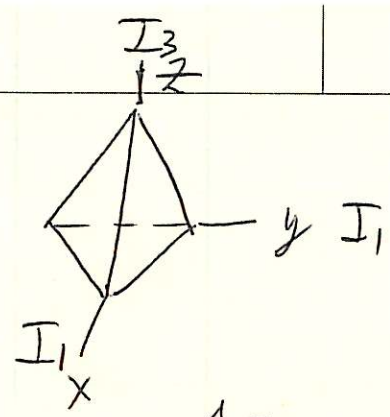
$$= \frac{1}{2} e^{i\hbar t/I} \left(-\frac{i}{\sqrt{2}} \right) \left(\langle 1,1| e^{i\omega_0 t} + \langle 1,-1| e^{-i\omega_0 t} \right) \cdot$$

$$\cdot (\hat{L}_+ + \hat{L}_-) \frac{i e^{-i\hbar t/I}}{\sqrt{2}} \left(e^{-i\omega_0 t} |1,1\rangle + e^{i\omega_0 t} |1,-1\rangle \right)$$

$$= \frac{1}{2\cdot\sqrt{2}} \left(\langle 1,1| e^{i\omega_0 t} + \langle 1,-1| e^{-i\omega_0 t} \right) \frac{\sqrt{2}\hbar}{\sqrt{2}} \left(e^{-i\omega_0 t} |1,0\rangle \right.$$

$$\left. + e^{i\omega_0 t} |1,0\rangle \right) = 0.$$

P9.21



(a) $$\hat{H} = \frac{\hat{L}_z^2}{2I_3} + \frac{\overbrace{\hat{L}_x^2 + \hat{L}_y^2}^{\hat{L}^2 - \hat{L}_z^2}}{2I_1} = \frac{\hat{L}^2}{2I_1} + \frac{\hat{L}_z^2}{2}\left(\frac{1}{I_3} - \frac{1}{I_1}\right)$$

(b) $[\hat{H}, \hat{L}_z] = 0$ because $[\hat{L}^2, \hat{L}_z] = 0$.

(c) The eigenstates are $|\ell, m\rangle$ states

$$\hat{H}\,|\ell, m\rangle = \left\{\frac{\hbar^2 \ell(\ell+1)}{2I_1} + \frac{\hbar^2 m^2}{2}\left(\frac{1}{I_3} - \frac{1}{I_1}\right)\right\}|\ell, m\rangle$$

$$E_{\ell,m} = \hbar^2\left(\frac{\ell(\ell+1) - m^2}{2I_1} + \frac{m^2}{2I_3}\right)$$

(d) $$|\psi(0)\rangle = \frac{1}{\sqrt{2}}\,|0,0\rangle + \frac{1}{\sqrt{2}}\,|1,1\rangle$$

$$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar}\,|\psi(0)\rangle =$$

$$= e^{-iE_{0,0}t/\hbar}\,\frac{|0,0\rangle}{\sqrt{2}} + e^{-iE_{1,1}t/\hbar}\,\frac{|1,1\rangle}{\sqrt{2}}$$

$$= \frac{1}{\sqrt{2}}\,|0,0\rangle + \frac{\exp\left[-i\hbar t\left(\frac{1}{2I_1} + \frac{1}{2I_3}\right)\right]}{\sqrt{2}}\,|1,1\rangle$$

$$E_{1,1} = \frac{\hbar^2}{2}\left(\frac{2-1}{I_1} + \frac{1}{I_3}\right) = \frac{\hbar^2}{2I_1} + \frac{\hbar^2}{2I_3}$$

$\boxed{P10.1}$ $\hat{p}_r \to \frac{\hbar}{i}\left(\frac{\partial}{\partial r} + \frac{1}{r}\right)$

$$\langle \psi | \hat{p}_r | \psi \rangle = \frac{\hbar}{i} \int d^3\vec{r}\; \psi^*(r,\theta,\phi)\left(\frac{\partial}{\partial r} + \frac{1}{r}\right)\psi(r,\theta,\phi)$$

$=$ (angular integration gives 1) $=$

$$= \frac{\hbar}{i} \int_0^\infty dr \cdot r^2 \, \frac{u^*}{r}\left(\frac{\partial}{\partial r} + \frac{1}{r}\right)\frac{u}{r} =$$

$$= \frac{\hbar}{i}\left\{\int_0^\infty dr \; r u^* \frac{\partial}{\partial r}\left(\frac{u}{r}\right) + \int_0^\infty dr \frac{|u|^2}{r}\right\} =$$

$$= \frac{\hbar}{i}\left\{\int_0^\infty dr \; r u^*\left(-\frac{u}{r^2} + \frac{1}{r}\frac{du}{dr}\right) + \int_0^\infty dr \frac{|u|^2}{r}\right\} =$$

$$= \frac{\hbar}{i}\left\{\int_0^\infty dr\left(-\frac{|u|^2}{r} + \frac{|u|^2}{r}\right) + \int_0^\infty dr \; u^* \frac{du}{dr}\right\}$$

$$= \frac{\hbar}{i}\left\{u^* u\Big|_0^\infty - \int_0^\infty dr \; u \frac{du^*}{dr}\right\} = \frac{\hbar}{i}\left\{-|u(0)|^2\right\}$$

$$- \frac{\hbar}{i}\int_0^\infty dr \; u \frac{du^*}{dr}$$

$$\langle \psi | \hat{p}_r | \psi \rangle^* = -\frac{\hbar}{i}\int_0^\infty dr \, r^2 \, \frac{u}{r}\left(\frac{\partial}{\partial r} + \frac{1}{r}\right)\frac{u^*}{r} =$$

$$= -\frac{\hbar}{i}\left\{\int_0^\infty dr \; \frac{u u^*}{r} + \int_0^\infty dr \; r u \; \frac{d(u^*/r)}{dr}\right\} =$$

$$= -\frac{\hbar}{i}\left\{\int_0^\infty dr \; \frac{|u|^2}{r} + \int dr \; r u\left(-\frac{u^*}{r^2} + \frac{1}{r}\frac{du^*}{dr}\right)\right\}$$

$$= -\frac{\hbar}{i}\int_0^\infty dr \; u \frac{du^*}{dr}$$

$\Rightarrow$ from $\langle \psi | \hat{p}_r | \psi \rangle = \langle \psi | \hat{p}_r | \psi \rangle^*$ follows that $|u(0)|^2 = 0$

$\boxed{P10.2}$ $|\psi\rangle = \frac{4}{5}|1,0,0\rangle + \frac{3i}{5}|2,1,1\rangle$

(a) $\hat{H}|1,0,0\rangle = E_1|1,0,0\rangle$; $\hat{H}|2,1,1\rangle = E_2|2,1,1\rangle$

where $E_n = -\frac{\mu c^2\alpha^2}{2n^2}$

$$\langle E\rangle = \langle\hat{H}\rangle = \frac{16}{25}\left(-\frac{\mu c^2\alpha^2}{2}\right) + \frac{9}{25}\left(-\frac{\mu c^2\alpha^2}{2\cdot 4}\right)$$

$$= -\frac{\mu c^2\alpha^2}{2}\,\frac{1}{25}\underbrace{\left(16+\frac{9}{4}\right)}_{\frac{73}{4}} = -\frac{73}{200}\mu c^2\alpha^2$$

$$\hat{L}^2|1,0,0\rangle = 0 = \hat{L}_z|1,0,0\rangle$$

$$\hat{L}^2|2,1,1\rangle = \hbar^2 1(1+1)|2,1,1\rangle = 2\hbar^2|2,1,1\rangle$$

$$\hat{L}_z|2,1,1\rangle = \hbar|2,1,1\rangle$$

$$\Rightarrow \langle\hat{L}^2\rangle = \frac{16}{25}(0) + \frac{9}{25}2\hbar^2 = \frac{18}{25}\hbar^2$$

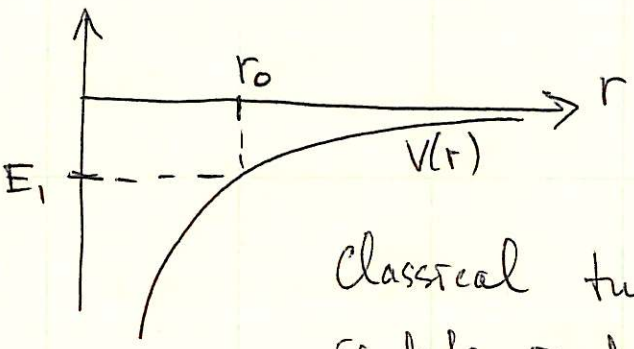
$$\langle\hat{L}_z\rangle = \frac{16}{25}(0) + \frac{9}{25}\hbar = \frac{9\hbar}{25}$$

(b) $|\psi(t)\rangle = e^{-i\hat{H}t/\hbar}|\psi(0)\rangle = e^{-iE_1t/\hbar}\frac{4}{5}|1,0,0\rangle +$

$$+ e^{-iE_2t/\hbar}\frac{3i}{5}|2,1,1\rangle$$

Since $\hat{H}$, $\hat{L}^2$ and $\hat{L}_z$ commute with Hamiltonian $\hat{H}$, their expectation values <u>do not</u> change with time.

$\boxed{P10.3}$ Ground state of hydrogen $E_1 = -\frac{\mu c^2 \alpha^2}{2}$ $(n=1, l=0=m)$



Classical turning point r_0 is determined by condition that kinetic energy $=0$ there

$$\Rightarrow \quad E_1 = \text{Pot. Energy} = -\frac{e^2}{r_0}$$

$$r_0 = -\frac{e^2}{E_1} = -\frac{e^2 \cdot 2}{-\mu c^2 \alpha^2} = 2\,\frac{e^2}{\mu c^2 \frac{e^4}{\hbar^2 c^2}} =$$

$$= 2\,\frac{\hbar^2 \cancel{c^2}}{\mu \cancel{c^2} e^2} = 2\,\frac{\hbar}{\mu c \alpha} = 2a_0$$

↳ Bohr radius

$\Rightarrow$ need to find probability of the electron being outside the $r_0 = 2a_0$ sphere:

$$P_> = \int_{2a_0}^{\infty} dr \cdot r^2 \left|R_{1,0}(r)\right|^2 = \int_{2a_0}^{\infty} dr \cdot r^2 \left(\frac{2}{a_0^{3/2}}\, e^{-r/a_0}\right)^2$$

$$= \frac{4}{a_0^3} \int_{2a_0}^{\infty} dr\, r^2 e^{-2r/a_0} = \left(\begin{array}{c} r = 2a_0 x \\ dr = 2a_0 dx \end{array}\right) =$$

$$= \frac{4}{a_0^3} (2a_0)^3 \int_1^{\infty} dx\, x^2 e^{-4x} = 32\left\{-\frac{1}{4}\int_1^{\infty} d\left(e^{-4x}\right) x^2\right\}$$

$$= 32\left\{-\frac{1}{4}\left(e^{-4x} x^2 \Big|_1^{\infty} - 2\int_1^{\infty} dx \cdot x\, e^{-4x}\right)\right\} = 32\left\{\frac{e^{-4}}{4} + \frac{1}{2}\cdot\left(-\frac{1}{4}\right)\cdot\right.$$

$$\left.\cdot \int_1^{\infty} d\left(e^{-4x}\right) x\right\} = 32\left\{\frac{e^{-4}}{4} - \frac{1}{8}\left(e^{-4x} x \Big|_1^{\infty} - \int_1^{\infty} dx\, e^{-4x}\right)\right\} = 32\, e^{-4}\left(\frac{1}{4} + \frac{1}{8} + \frac{1}{32}\right)$$

$$= 13\, e^{-4} = 0.24$$